

Requirement Specification:

TRUST Tech Deliverable 4a: Enable user GitLab & Docker images

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Contents

1 Design Specification: TRUST Tech Deliverable 4a. Enable user GitLab & Docker images	3
2 Document Title	3
2.1 Table of Contents	3
2.2 1. Introduction	5
2.2.1 1.1 References	5
2.3 2. Application Overview	5
2.3.1 2.1 Layers or architectural framework	5
2.3.2 2.2 Deployment View	5
2.3.3 2.3 Development View	5
2.3.4 2.4 Architectural Mechanisms	5
2.4 3. Architecturally significant requirements	5
2.5 4. Security	5
2.5.1 4.1 System Security	5
2.5.2 4.2 Data Security	6
2.5.3 4.3 Application Security	6
2.6 5. Common Modules	6
2.7 6. Functional Modules	6
2.8 7. Message / Service Interface	6
2.9 8. Database Design	6
2.10 Appendix A	6

1 Design Specification: TRUST Tech Deliverable 4a. Enable user GitLab & Docker images

2 Document Title

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Revision History NA

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2.1 Table of Contents

1. Introduction
 - 1.1 References
2. Application Overview
 - 2.1 Layers or architectural framework
 - 2.2 Deployment View
 - 2.3 Development View
 - 2.4 Architectural Mechanisms
3. Architecturally significant requirements
4. Security
 - 4.1 System Security
 - 4.2 Data Security
 - 4.3 Application Security
5. Common Modules: NA
6. Functional Modules: NA
7. Message / Service Interface: NA

8. Database Design: NA

- Appendix A: NA
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2.2 1. Introduction

2.2.1 1.1 References

- GovTech GCC Security Guidelines
 - AWS Best Practices for Multi-Account Strategy
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2.3 2. Application Overview

2.3.1 2.1 Layers or architectural framework

The solution follows a multi-account AWS architecture for hosting CloudOS and related infrastructure components:

- AWS Service Account: Hosts CloudOS and GitLab Server for private code repositories.
- AWS Workload Account: Hosts AWS Batch and AWS ECR for containerized workload execution.

2.3.2 2.2 Deployment View

- GitLab Server: Deployed in the AWS Service Account, configured for private access.
- AWS ECR: Deployed in the AWS Workload Account, accessible only within AWS GCC accounts.
- AWS Batch: Utilized for running containerized jobs, with required IAM permissions.

2.3.3 2.3 Development View

- GitLab provides private repository hosting for researchers and engineers.
- CI/CD workflows will be managed within GitLab to integrate with AWS Batch.
- AWS ECR will store and manage private container images for workloads.

2.3.4 2.4 Architectural Mechanisms

- Secure networking between AWS accounts using VPC peering or AWS PrivateLink.
 - IAM roles and policies restricting access to only necessary resources.
-

2.4 3. Architecturally significant requirements

- Deployment in a non-internet environment within Singapore GCC.
 - Secure and private access for all services.
 - Multi-account strategy for isolation and compliance.
-

2.5 4. Security

2.5.1 4.1 System Security

- Network access controls enforced via Security Groups and VPC.
- AWS KMS for encrypting sensitive data and images.

2.5.2 4.2 Data Security

- No external data access permitted; all data resides within AWS GCC.
- Private container images stored in AWS ECR with strict access policies.

2.5.3 4.3 Application Security

- GitLab access restricted to authorized users only.
 - IAM roles configured for least privilege access.
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2.6 5. Common Modules

- NA
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2.7 6. Functional Modules

- NA
-

2.8 7. Message / Service Interface

- NA
-

2.9 8. Database Design

- NA
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2.10 Appendix A

- NA